

MH1201 Linear Algebra II

Midterm Exam (Solutions)

AY 2022/2023, Semester 2

Quantitative Research Society @NTU

August 17, 2026

Overview

- Linear maps on matrix spaces; matrix representations w.r.t. standard bases; null space and range; rank–nullity.
- Subspace tests and counterexamples in function spaces, including polynomial-degree constraints and exponential–polynomial classes.
- Subspace criteria: closure under addition/scalar multiplication vs. nonemptiness; equivalences via standard subspace tests.
- Solution structure: each question is solved using multiple independent methods (Method 1 explanatory, Method 2 compact/symbolic, and an additional method when natural).

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Question 1**(15 marks)****Problem**

Let $T : \mathcal{M}_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^4$ be the map defined by

$$T\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) = (a_{11} + a_{12}, a_{21} + a_{22}, a_{11} + a_{21}, a_{12} + a_{22}).$$

- (a) Show that T is a linear map.
- (b) Let $\mathcal{B} = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ be the standard basis for $\mathcal{M}_{2 \times 2}(\mathbb{R})$ and $\mathcal{A} = \{e_1, e_2, e_3, e_4\}$ be the standard basis for $\mathbb{R}^4$. Write down the matrix representation of T with respect to $\mathcal{B}$ and $\mathcal{A}$.
- (c) Determine a basis for either $\text{null}(T)$ OR $\text{range}(T)$.
- (d) Determine the dimensions of $\text{null}(T)$ AND $\text{range}(T)$.
- (e) Determine whether the following properties of T is true. You are to explain your answer.
 - (i) Is T injective?
 - (ii) Is T surjective?

Solution**Method 1: Coordinate computation + row reduction (standard)**

Write a general matrix as $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, so

$$T(A) = (a + b, c + d, a + c, b + d).$$

- (a) **Linearity.** Let $A, B \in \mathcal{M}_{2 \times 2}(\mathbb{R})$ and $\lambda \in \mathbb{R}$. Writing $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$, we have

$$\begin{aligned} T(A + B) &= T\left(\begin{bmatrix} a + a' & b + b' \\ c + c' & d + d' \end{bmatrix}\right) \\ &= (a + a' + b + b', c + c' + d + d', a + a' + c + c', b + b' + d + d') \\ &= T(A) + T(B), \end{aligned}$$

and

$$T(\lambda A) = T\left(\begin{bmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{bmatrix}\right) = (\lambda(a + b), \lambda(c + d), \lambda(a + c), \lambda(b + d)) = \lambda T(A).$$

Hence T is linear.

(b) **Matrix representation.** Compute T on the standard basis $\mathcal{B} = (E_{11}, E_{12}, E_{21}, E_{22})$:

$$T(E_{11}) = (1, 0, 1, 0), \quad T(E_{12}) = (1, 0, 0, 1), \quad T(E_{21}) = (0, 1, 1, 0), \quad T(E_{22}) = (0, 1, 0, 1).$$

Since $\mathcal{A}$ is the standard basis, the j -th column of $[T]_{\mathcal{B}}^{\mathcal{A}}$ is $[T(E_{.j})]_{\mathcal{A}} = T(E_{.j})$. Therefore

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

(c) **A basis for $\text{null}(T)$.** Solve $T(A) = 0$ for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. The system is

$$a + b = 0, \quad c + d = 0, \quad a + c = 0, \quad b + d = 0.$$

From $a + b = 0$, $b = -a$; from $a + c = 0$, $c = -a$; then $c + d = 0$ gives $d = a$. Thus

$$\text{null}(T) = \left\{ a \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} : a \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right\}.$$

$$\text{A basis for } \text{null}(T) \text{ is } \left\{ \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right\}.$$

(d) **Dimensions of $\text{null}(T)$ and $\text{range}(T)$.** From (c), $\dim \text{null}(T) = 1$.

Row-reduce $[T]_{\mathcal{B}}^{\mathcal{A}}$:

$$\begin{aligned} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} &\xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} &\xrightarrow{R_3 \leftarrow -R_3} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} &\xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} &\xrightarrow{R_2 \leftarrow R_2 - R_4} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}. \end{aligned}$$

There are 3 pivot rows, hence $\text{rank}(T) = 3$ and $\dim \text{range}(T) = 3$. (Check: $1 + 3 = 4$.)

$$\dim \text{null}(T) = 1, \quad \dim \text{range}(T) = 3.$$

(e) **Injective / surjective.**

(i) T is injective $\iff \text{null}(T) = \{0\}$. Since $\dim \text{null}(T) = 1$, T is not injective.

(ii) T is surjective $\iff \text{range}(T) = \mathbb{R}^4$. Since $\dim \text{range}(T) = 3 < 4$, T is not surjective.

T is neither injective nor surjective.

Method 2: Rank via a linear relation + a nonzero minor (compact)

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and set

$$T(A) = (a + b, c + d, a + c, b + d) =: (y_1, y_2, y_3, y_4).$$

- (a) **Linearity.** Each component $A \mapsto a + b$, $A \mapsto c + d$, $A \mapsto a + c$, $A \mapsto b + d$ is a linear functional on $\mathcal{M}_{2 \times 2}(\mathbb{R})$, hence T is linear.
- (b) **Matrix representation.** As in Method 1,

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

- (c) **Basis for range(T).** Let

$$v_1 = T(E_{11}) = (1, 0, 1, 0), \quad v_2 = T(E_{12}) = (1, 0, 0, 1), \quad v_3 = T(E_{21}) = (0, 1, 1, 0).$$

Consider the 3×3 minor using rows 1, 2, 3:

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = -(0 \cdot 1 - 1 \cdot 1) = 1 \neq 0.$$

Hence v_1, v_2, v_3 are linearly independent, so $\dim \text{range}(T) \geq 3$.

$$\text{A basis for } \text{range}(T) \text{ is } \{(1, 0, 1, 0), (1, 0, 0, 1), (0, 1, 1, 0)\}.$$

- (d) **Dimensions.** For any A ,

$$y_1 + y_2 = (a + b) + (c + d) = a + b + c + d = (a + c) + (b + d) = y_3 + y_4,$$

so $\text{range}(T) \subseteq H := \{y \in \mathbb{R}^4 : y_1 + y_2 - y_3 - y_4 = 0\}$ and $\dim H = 3$. Thus $\dim \text{range}(T) \leq 3$, and together with (c) we get $\dim \text{range}(T) = 3$.

By the Rank–Nullity Theorem,

$$\dim \text{null}(T) = 4 - \dim \text{range}(T) = 4 - 3 = 1.$$

$$\dim \text{null}(T) = 1, \quad \dim \text{range}(T) = 3.$$

- (e) **Injective / surjective.**

- (i) $\dim \text{null}(T) = 1 \neq 0 \Rightarrow T$ not injective.
- (ii) $\dim \text{range}(T) = 3 < 4 \Rightarrow T$ not surjective.

$$T \text{ is neither injective nor surjective.}$$

Method 3: Row/column-sum interpretation

Interpret

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (r_1, r_2, c_1, c_2),$$

where $r_1 = a + b$, $r_2 = c + d$ are the row sums and $c_1 = a + c$, $c_2 = b + d$ are the column sums.

(a) **Linearity.** Row sums and column sums are linear in the entries, hence T is linear.

(b) **Matrix representation.** Using $T(E_{ij})$ as in Method 1 yields

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

(c) **Basis for $\text{range}(T)$.** Every output satisfies

$$r_1 + r_2 = (a + b) + (c + d) = a + b + c + d = (a + c) + (b + d) = c_1 + c_2,$$

so $\text{range}(T) \subseteq \{(r_1, r_2, c_1, c_2) : r_1 + r_2 = c_1 + c_2\}$. Conversely, given (r_1, r_2, c_1, c_2) with $r_1 + r_2 = c_1 + c_2$, one can solve

$$a + b = r_1, \quad c + d = r_2, \quad a + c = c_1, \quad b + d = c_2$$

(consistency is exactly $r_1 + r_2 = c_1 + c_2$).

Hence

$$\text{range}(T) = \{y \in \mathbb{R}^4 : y_1 + y_2 = y_3 + y_4\},$$

a 3-dimensional subspace. A convenient basis is

$$(1, 0, 1, 0), \quad (0, 1, 0, 1), \quad (1, 0, 0, 1).$$

$$\boxed{\text{A basis for } \text{range}(T) \text{ is } \{(1, 0, 1, 0), (0, 1, 0, 1), (1, 0, 0, 1)\}.$$

(d) **Dimensions.** From (c), $\dim \text{range}(T) = 3$, hence by rank–nullity $\dim \text{null}(T) = 4 - 3 = 1$.

$$\boxed{\dim \text{null}(T) = 1, \quad \dim \text{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) $\dim \text{null}(T) = 1 \Rightarrow T$ not injective.

(ii) $\text{range}(T)$ is a proper subspace of $\mathbb{R}^4$ (e.g. $(1, 0, 0, 0) \notin \text{range}(T)$ since $1 + 0 \neq 0 + 0$), hence T not surjective.

$$\boxed{T \text{ is neither injective nor surjective.}}$$

Method 4: Operator factorisation via left/right multiplication

Let $u = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{R}^2$. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we have

$$Au = \begin{bmatrix} a+b \\ c+d \end{bmatrix}, \quad u^T A = \begin{bmatrix} a+c & b+d \end{bmatrix}.$$

Hence, identifying $\mathbb{R}^2 \times \mathbb{R}^2 \cong \mathbb{R}^4$ via (x_1, x_2, y_1, y_2) ,

$$T(A) = (Au, u^T A) = (a+b, c+d, a+c, b+d).$$

(a) **Linearity.** The map $L : \mathcal{M}_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^2$, $L(A) = Au$, is linear since $(A+B)u = Au + Bu$ and $(\lambda A)u = \lambda(Au)$. Similarly, $R : \mathcal{M}_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^2$, $R(A) = u^T A$, is linear since $u^T(A+B) = u^T A + u^T B$ and $u^T(\lambda A) = \lambda u^T A$. Therefore $T(A) = (L(A), R(A))$ is linear.

(b) **Matrix representation.** Compute $T(E_{ij})$ (as in Method 1):

$$T(E_{11}) = (1, 0, 1, 0), \quad T(E_{12}) = (1, 0, 0, 1), \quad T(E_{21}) = (0, 1, 1, 0), \quad T(E_{22}) = (0, 1, 0, 1),$$

so

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

(c) **A basis for $\text{null}(T)$.** We have $T(A) = 0$ iff $Au = 0$ and $u^T A = 0$. Writing $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the conditions $Au = 0$ give

$$a+b=0, \quad c+d=0,$$

and $u^T A = 0$ gives

$$a+c=0, \quad b+d=0.$$

Solving: $b = -a$, $c = -a$, $d = a$, hence

$$\text{null}(T) = \left\{ a \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} : a \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right\}.$$

$$\text{A basis for } \text{null}(T) \text{ is } \left\{ \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right\}.$$

(d) **Dimensions.** From (c), $\dim \text{null}(T) = 1$.

Also, for any A , the total sum of entries equals both the sum of row sums and the sum of column sums:

$$(a+b) + (c+d) = (a+c) + (b+d),$$

so $\text{range}(T) \subseteq H := \{y \in \mathbb{R}^4 : y_1 + y_2 = y_3 + y_4\}$ and $\dim H = 3$. Conversely, given $y = (y_1, y_2, y_3, y_4) \in H$ (so $y_1 + y_2 = y_3 + y_4$), choose any $a \in \mathbb{R}$ and set

$$b = y_1 - a, \quad c = y_3 - a, \quad d = y_4 - b = y_4 - (y_1 - a) = a + y_4 - y_1.$$

Then $a+b = y_1$, $a+c = y_3$, $b+d = y_4$, and

$$c+d = (y_3 - a) + (a + y_4 - y_1) = y_3 + y_4 - y_1 = y_2$$

(using $y_1 + y_2 = y_3 + y_4$). Hence $T(A) = y$ and $\text{range}(T) = H$, so $\dim \text{range}(T) = 3$.

$$\dim \text{null}(T) = 1, \quad \dim \text{range}(T) = 3.$$

(e) **Injective / surjective.**

(i) Since $\dim \text{null}(T) = 1 \neq 0$, T is not injective.

(ii) Since $\text{range}(T) = H \neq \mathbb{R}^4$ (a proper hyperplane), T is not surjective.

T is neither injective nor surjective.

Method 5: Hyperplane characterisation + First Isomorphism Theorem

Theorem (First Isomorphism Theorem). If $S : V \rightarrow W$ is a linear map, then $V/\ker(S) \cong \text{range}(S)$. In particular, if V is finite-dimensional, then

$$\dim V = \dim \ker(S) + \dim \text{range}(S).$$

Define the hyperplane

$$H := \{y = (y_1, y_2, y_3, y_4) \in \mathbb{R}^4 : y_1 + y_2 - y_3 - y_4 = 0\}.$$

- (a) **Linearity.** As in Method 1, each component of $T(A) = (a+b, c+d, a+c, b+d)$ is a linear functional in the entries of A , so T is linear.
- (b) **Matrix representation.** Compute T on $\mathcal{B} = (E_{11}, E_{12}, E_{21}, E_{22})$:

$$T(E_{11}) = (1, 0, 1, 0), \quad T(E_{12}) = (1, 0, 0, 1), \quad T(E_{21}) = (0, 1, 1, 0), \quad T(E_{22}) = (0, 1, 0, 1),$$

hence

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

- (c) **A basis for $\text{range}(T)$.** First show $\text{range}(T) \subseteq H$: for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $T(A) = (y_1, y_2, y_3, y_4)$,

$$y_1 + y_2 - y_3 - y_4 = (a+b) + (c+d) - (a+c) - (b+d) = 0,$$

so $T(A) \in H$.

Next show $H \subseteq \text{range}(T)$: take any $y = (y_1, y_2, y_3, y_4) \in H$ so $y_1 + y_2 = y_3 + y_4$. Choose any $a \in \mathbb{R}$ and define

$$b = y_1 - a, \quad c = y_3 - a, \quad d = y_4 - b.$$

Then $a+b = y_1$, $a+c = y_3$, $b+d = y_4$, and

$$c+d = (y_3 - a) + d = (y_3 - a) + (y_4 - b) = y_3 + y_4 - (a+b) = y_3 + y_4 - y_1 = y_2$$

(using $y_1 + y_2 = y_3 + y_4$). Hence $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = y$. Therefore $\text{range}(T) = H$.

A convenient basis for H is, for instance,

$$h_1 = (1, 0, 1, 0), \quad h_2 = (0, 1, 0, 1), \quad h_3 = (1, 0, 0, 1),$$

which lie in H and are linearly independent. Thus they form a basis of $\text{range}(T) = H$.

$$\boxed{\text{A basis for } \text{range}(T) \text{ is } \{(1, 0, 1, 0), (0, 1, 0, 1), (1, 0, 0, 1)\}.$$

- (d) **Dimensions via the First Isomorphism Theorem.** Since $\text{range}(T) = H$ and $\dim H = 3$, we have $\dim \text{range}(T) = 3$. Because $\dim \mathcal{M}_{2 \times 2}(\mathbb{R}) = 4$, the First Isomorphism Theorem (dimension form) gives

$$4 = \dim \text{null}(T) + 3 \quad \Rightarrow \quad \dim \text{null}(T) = 1.$$

$$\boxed{\dim \text{null}(T) = 1, \quad \dim \text{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) T is injective $\iff \text{null}(T) = \{0\}$. Since $\dim \text{null}(T) = 1$, T is not injective.

(ii) T is surjective $\iff \text{range}(T) = \mathbb{R}^4$. Since $\text{range}(T) = H \neq \mathbb{R}^4$, T is not surjective.

T is neither injective nor surjective.

Mark Scheme (15 marks)• **(a) (2 marks)**

- (1) Shows additivity: $T(A + B) = T(A) + T(B)$ with correct componentwise evaluation.
- (1) Shows homogeneity: $T(\lambda A) = \lambda T(A)$ with correct componentwise evaluation.

• **(b) (4 marks)**

- (1) Correctly computes $T(E_{11})$.
- (1) Correctly computes $T(E_{12})$.
- (1) Correctly computes $T(E_{21})$ and $T(E_{22})$.
- (1) Assembles $[T]_{\mathcal{B}}^{\mathcal{A}}$ with correct column order matching $\mathcal{B}$ and standard coordinates in $\mathcal{A}$.

• **(c) (3 marks)**

- (1) Sets up the condition defining $\text{null}(T)$ or $\text{range}(T)$ (e.g. $T(A) = 0$ or a correct range constraint).
- (1) Solves / constructs correctly to obtain a spanning set for the requested space.
- (1) States a valid basis for either $\text{null}(T)$ or $\text{range}(T)$ and indicates why it spans (and is independent if required).

• **(d) (2 marks)**

- (1) Correctly finds $\dim \text{range}(T)$ (via row-reduction, minor, or a correct dimension argument).
- (1) Correctly finds $\dim \text{null}(T)$ (via basis from (c) and/or rank-nullity / isomorphism theorem) and states both dimensions.

• **(e) (4 marks)**

- (2) Injectivity: uses criterion T injective $\iff \text{null}(T) = \{0\}$ (or $\dim(T) = 0$) and gives correct conclusion.
- (2) Surjectivity: uses criterion T surjective $\iff \text{range}(T) = \mathbb{R}^4$ (or $\dim \text{range}(T) = 4$) and gives correct conclusion.

Question 2**(10 marks)****Problem**

Recall that $\mathcal{F}(\mathbb{R}, \mathbb{R})$ is the vector space comprising all functions with domain $\mathbb{R}$ and codomain $\mathbb{R}$.

- (a) Consider the set $\mathcal{U} = \{f \in \mathcal{P}(\mathbb{R}) : \deg(f) \geq 2023\}$. Is $\mathcal{U}$ a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$?
 (b) Consider the set $\mathcal{V} = \{e^x f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) : f \in \mathcal{P}(\mathbb{R})\}$. Is $\mathcal{V}$ a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$?

Solution**Method 1: Subspace test**

- (a) **$\mathcal{U}$ is not a subspace.** If $\mathcal{U}$ were a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$, it must contain the zero vector (the zero function, i.e. the zero polynomial). Pick $f(x) = x^{2023} \in \mathcal{U}$. Closure under scalar multiplication would imply $0 \cdot f \in \mathcal{U}$. But $0 \cdot f = 0$ is the zero polynomial, whose degree is not ≥ 2023 (indeed $\deg(0)$ is undefined; under the standard convention one may take $\deg(0) = -\infty$). Hence $0 \notin \mathcal{U}$, so $\mathcal{U}$ is not a subspace.

 $\mathcal{U}$ is not a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

- (b) **$\mathcal{V}$ is a subspace.** First, $0 \in \mathcal{V}$ since taking $f = 0 \in \mathcal{P}(\mathbb{R})$ gives $e^x f = 0$.
 Let $e^x f, e^x g \in \mathcal{V}$ with $f, g \in \mathcal{P}(\mathbb{R})$ and let $\lambda \in \mathbb{R}$. Then

$$(e^x f) + (e^x g) = e^x(f + g) \in \mathcal{V} \quad \text{since } f + g \in \mathcal{P}(\mathbb{R}),$$

and

$$\lambda(e^x f) = e^x(\lambda f) \in \mathcal{V} \quad \text{since } \lambda f \in \mathcal{P}(\mathbb{R}).$$

Thus $\mathcal{V}$ is nonempty and closed under addition and scalar multiplication, hence a subspace.

 $\mathcal{V}$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.
Method 2: Range of a linear map is a subspace (structural)

Theorem. If $\Phi : V \rightarrow W$ is linear, then $\text{range}(\Phi)$ is a subspace of W .

Define

$$\Phi : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{F}(\mathbb{R}, \mathbb{R}), \quad \Phi(f) = e^x f.$$

- (a) If $\mathcal{U}$ were a subspace then $0 \in \mathcal{U}$. But $0 \notin \mathcal{U}$ (degree constraint), so $\mathcal{U}$ is not a subspace.

 $\mathcal{U}$ is not a subspace.

- (b) Φ is linear since $\Phi(f + g) = e^x(f + g) = e^x f + e^x g$ and $\Phi(\lambda f) = e^x(\lambda f) = \lambda e^x f$. Moreover, by definition,

$$\text{range}(\Phi) = \{e^x f : f \in \mathcal{P}(\mathbb{R})\} = \mathcal{V}.$$

Hence $\mathcal{V} = \text{range}(\Phi)$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

 $\mathcal{V}$ is a subspace.

Method 3: Multiplication operator is a linear automorphism (operator-theoretic)

Define the multiplication operator

$$M_{e^x} : \mathcal{F}(\mathbb{R}, \mathbb{R}) \rightarrow \mathcal{F}(\mathbb{R}, \mathbb{R}), \quad (M_{e^x}h)(x) = e^x h(x).$$

- (a) As before, $0 \notin \mathcal{U}$ (since $\deg(0)$ is not ≥ 2023), so $\mathcal{U}$ is not a subspace.

$\mathcal{U}$ is not a subspace.

- (b) M_{e^x} is linear. Since $e^x \neq 0$ for all $x \in \mathbb{R}$, the inverse operator is

$$M_{e^{-x}} : \mathcal{F}(\mathbb{R}, \mathbb{R}) \rightarrow \mathcal{F}(\mathbb{R}, \mathbb{R}), \quad (M_{e^{-x}}h)(x) = e^{-x}h(x),$$

and it satisfies $M_{e^{-x}} \circ M_{e^x} = M_{e^x} \circ M_{e^{-x}} = \text{Id}$. Thus M_{e^x} is a linear isomorphism of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Now $\mathcal{P}(\mathbb{R})$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$, and

$$\mathcal{V} = \{e^x f : f \in \mathcal{P}(\mathbb{R})\} = M_{e^x}(\mathcal{P}(\mathbb{R})).$$

The image of a subspace under a linear map is a subspace; in particular, the image under an isomorphism is a subspace. Hence $\mathcal{V}$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

$\mathcal{V}$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Method 4: Subspace isomorphism viewpoint (explicit inverse on $\mathcal{V}$)

Define a map

$$\psi : \mathcal{V} \rightarrow \mathcal{P}(\mathbb{R}), \quad \psi(e^x f) = f.$$

This is well-defined because if $e^x f = e^x g$, then multiplying by e^{-x} (pointwise) gives $f = g$.

- (a) As above, $0 \notin \mathcal{U}$, so $\mathcal{U}$ is not a subspace.

$\mathcal{U}$ is not a subspace.

- (b) Consider the restriction $\phi := M_{e^x}|_{\mathcal{P}(\mathbb{R})} : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{V}$ given by $\phi(f) = e^x f$. Then ϕ is linear and bijective with inverse ψ (since $\psi(\phi(f)) = f$ and $\phi(\psi(e^x f)) = e^x f$). Thus ϕ is a vector space isomorphism $\mathcal{P}(\mathbb{R}) \cong \mathcal{V}$. Since $\mathcal{P}(\mathbb{R})$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$, it follows that $\mathcal{V}$ is a subspace as well (equivalently: $\mathcal{V}$ is the image of a subspace under a linear map).

$\mathcal{V}$ is a subspace.

Mark Scheme (10 marks)**• (a) (5 marks)**

- (1) States a necessary subspace condition to test (e.g. $0 \in \mathcal{U}$, or closure under scalar multiplication).
- (1) Provides a valid example $f \in \mathcal{U}$ (e.g. $f(x) = x^{2023}$).
- (1) Uses closure under scalar multiplication to infer $0 \cdot f \in \mathcal{U}$ would be required if $\mathcal{U}$ were a subspace.
- (1) Observes $0 \cdot f = 0$ and explains why $0 \notin \mathcal{U}$ (degree not ≥ 2023 ; $\deg(0)$ undefined or $-\infty$ by convention).
- (1) Concludes correctly that $\mathcal{U}$ is not a subspace.

• (b) (5 marks)

- (1) Shows $0 \in \mathcal{V}$ (take $f = 0$).
- (2) Closure under addition: $(e^x f) + (e^x g) = e^x(f + g)$ and $f + g \in \mathcal{P}(\mathbb{R})$.
- (1) Closure under scalar multiplication: $\lambda(e^x f) = e^x(\lambda f)$ and $\lambda f \in \mathcal{P}(\mathbb{R})$.
- (1) Concludes correctly that $\mathcal{V}$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Question 3**(5 marks)****Problem**

Let V be a vector space. Let S be a subset of V that is closed under vector addition and scalar multiplication. Show that S is a subspace of V if and only if S is nonempty.

Solution**Method 1: Direct subspace test**

($\Rightarrow$) If S is a subspace of V , then by definition $0 \in S$. In particular, S is nonempty.

($\Leftarrow$) Suppose S is nonempty. Choose $s \in S$. Since S is closed under scalar multiplication, $0 \cdot s \in S$, so $0 \in S$. Now let $u, v \in S$ and $\lambda \in \mathbb{R}$. Since S is closed under addition and scalar multiplication, we have $u + v \in S$ and $\lambda u \in S$. Hence S contains 0 and is closed under addition and scalar multiplication, so S is a subspace of V .

$$S \text{ is a subspace of } V \iff S \neq \emptyset.$$

Method 2: Linear combinations criterion

Theorem (Linear combinations subspace test). A subset $S \subseteq V$ is a subspace iff $S \neq \emptyset$ and for all $u, v \in S$ and $\alpha, \beta \in \mathbb{R}$,

$$\alpha u + \beta v \in S.$$

Under the assumptions, if $S \neq \emptyset$ then for $u, v \in S$ and $\alpha, \beta \in \mathbb{R}$ we have $\alpha u \in S$ and $\beta v \in S$ by scalar closure, hence $\alpha u + \beta v \in S$ by additive closure. Therefore S is a subspace. Conversely, any subspace is nonempty.

$$S \leq V \iff S \neq \emptyset.$$

Method 3: Additive subgroup criterion

Theorem. A subset $S \subseteq V$ is a subspace iff: (i) S is a subgroup of $(V, +)$ and (ii) S is closed under scalar multiplication.

Assume $S \neq \emptyset$ and is closed under addition and scalar multiplication. Pick $s \in S$. Then $0 = 0 \cdot s \in S$. For any $u \in S$, we have $-u = (-1)u \in S$, so S is closed under additive inverses. Together with closure under addition, S is a subgroup of $(V, +)$; with scalar closure, S is a subspace. The converse direction is immediate since any subspace contains 0 and is nonempty.

$$S \text{ is a subspace of } V \iff S \neq \emptyset.$$

Method 4: Minimalist closure argument

($\Rightarrow$) If S is a subspace, then $0 \in S$, so $S \neq \emptyset$.

($\Leftarrow$) Suppose $S \neq \emptyset$ and fix $s \in S$. Then $0 = 0 \cdot s \in S$ by scalar closure. For any $u \in S$, $-u = (-1)u \in S$, and hence for any $u, v \in S$ we have $u - v = u + (-v) \in S$. Thus S contains 0 and is closed under addition and scalar multiplication, so S is a subspace.

$$S \leq V \iff S \neq \emptyset.$$

Mark Scheme (5 marks)**• (5 marks)**

- (1) Proves ($\Rightarrow$) direction: subspace $\Rightarrow 0 \in S \Rightarrow S \neq \emptyset$.
- (1) For ($\Leftarrow$) direction, assumes $S \neq \emptyset$ and selects $s \in S$.
- (1) Uses scalar closure to derive $0 = 0 \cdot s \in S$.
- (1) Uses closure (addition and scalar multiplication) to verify subspace axioms for arbitrary elements (or derives additive inverses and closure under linear combinations).
- (1) States the final equivalence clearly.